Quantification: 682_z29
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 15 |
| interior lacuna count | 13 |
| frame-edge lacuna count | 2 |

Interior lacunae only: mean, median and sample SD (n = 13)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1650.75 | 1511.5 | 948.377 | px^2 |
| major_axis_length_px | lacuna shape | 74.0009 | 74.236 | 22.3194 | px |
| minor_axis_length_px | lacuna shape | 27.9394 | 29.9101 | 9.5919 | px |
| aspect_ratio | lacuna shape | 2.805 | 2.6218 | 0.775 | unitless |
| eccentricity | lacuna shape | 0.9144 | 0.9241 | 0.0615 | unitless |
| solidity | lacuna shape | 0.8593 | 0.8693 | 0.0551 | unitless |
| roots_count | headline | 12.6154 | 14 | 4.5376 | count |
| ring_length_r30_px | headline | 343.692 | 363 | 128.306 | px |
| ring_length_r60_px | reported | 824.077 | 938 | 287.943 | px |
| owned_length_px | ownership-dependent | 2440.89 | 1579.88 | 1838.77 | px |
| edge_count | ownership-dependent | 106.846 | 62 | 83.2715 | count |
| mean_edge_length_px | ownership-dependent | 23.4937 | 23.5198 | 2.0628 | px |
| perimeter_px | normalised | 186.054 | 202.693 | 70.3766 | px |
| ring_area_r30_px2 | normalised | 7571.77 | 7711 | 2170.77 | px^2 |
| ring_area_r60_px2 | normalised | 19073 | 19480 | 4883.71 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9952 | 1 | 0.0173 | unitless |
| in_frame_fraction_r60 | normalised | 0.9839 | 1 | 0.0537 | unitless |
| ring_density_r30 | normalised | 0.0453797 | 0.0442874 | 0.0106587 | px^-1 |
| ring_density_r60 | normalised | 0.0429315 | 0.0434665 | 0.00848081 | px^-1 |
| roots_per_100px_perimeter | normalised | 7.1329 | 6.4136 | 2.0161 | per 100 px |
| ring_attached_length_r30_px | network v2 | 311.923 | 319 | 128.069 | px |
| ring_attached_length_r60_px | network v2 | 709.769 | 739 | 295.661 | px |
| ring_length_w_r30_px | network v2 | 386.788 | 410.635 | 141 | px |
| ring_length_w_r60_px | network v2 | 929.157 | 1048.9 | 323.319 | px |
| sholl_crossings_r10 | network v2 | 8.2308 | 8 | 3.0319 | count |
| sholl_crossings_r20 | network v2 | 10.5385 | 11 | 4.0541 | count |
| sholl_crossings_r30 | network v2 | 11.6923 | 11 | 3.9872 | count |
| loop_count | loops, report only | 10.3846 | 3 | 13.2636 | count |
| loop_edge_share | loops, report only | 0.2305 | 0.2256 | 0.2031 | share |
| ring_width_mean_r30_px | width | 6.9699 | 6.9555 | 0.6104 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02273e+06 | px^2 |
| total_skeleton_length_px | 38267 | px |
| canalicular_length_density_per_px | 0.0374166 | px^-1 |
| junction_count | 855 | count |
| junction_density_per_px2 | 0.000836 | px^-2 |
| lacuna_count | 15 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.467e-05 | px^-2 |
| skeleton_component_count | 230 | count |
| mean_component_length_px | 166.378 | px |
| median_component_length_px | 21 | px |
| field_length_density_w_per_px | 0.0422627 | px^-1 |
| field_density_without_flagged_per_px | 0.0377562 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 419 | px |
| unattached_component_count | 27 | count |
| canalicular_length_density_attached_per_px | 0.0374166 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0378263 | px^-1 |
| loop_count | 172 | count |
| loop_edge_share | 0.3531 | share |
| edge_count | 1869 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.7082 | px |
| width_median_px | 8 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 35757 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2083 | 86.508 | 34.896 | 2.479 | 0.915 | 0.8266 | 1.5435 | 13.4983 | 99.4023 | True |
| 2 | 2213 | 89.1614 | 32.1486 | 2.7734 | 0.9327 | 0.8778 | -1.1291 | 134.914 | 921.292 | False |
| 3 | 446 | 49.6136 | 12.2658 | 4.0449 | 0.969 | 0.7866 | -1.4445 | 249.987 | 449.359 | False |
| 4 | 3140 | 120.617 | 34.4194 | 3.5043 | 0.9584 | 0.8885 | 1.5609 | 431.048 | 220.093 | False |
| 5 | 2837 | 88.9183 | 41.496 | 2.1428 | 0.8844 | 0.9229 | -1.5453 | 532.043 | 493.055 | False |
| 6 | 1634 | 75.9278 | 30.0551 | 2.5263 | 0.9183 | 0.8211 | -1.4797 | 555.483 | 357.187 | False |
| 7 | 3781 | 121.501 | 43.6088 | 2.7862 | 0.9334 | 0.8771 | 1.4272 | 590.482 | 55.0987 | True |
| 8 | 1399 | 81.52 | 23.1918 | 3.515 | 0.9587 | 0.8541 | 1.5542 | 716.135 | 181.57 | False |
| 9 | 663 | 61.7115 | 15.8655 | 3.8897 | 0.9664 | 0.7408 | -1.1603 | 728.596 | 861.06 | False |
| 10 | 513 | 41.8409 | 17.3844 | 2.4068 | 0.9096 | 0.8607 | 1.3405 | 830.565 | 75.5867 | False |
| 11 | 1130 | 64.3164 | 23.6688 | 2.7173 | 0.9298 | 0.8835 | 1.4062 | 842.292 | 125.237 | False |
| 12 | 1244 | 49.7173 | 33.3792 | 1.4895 | 0.7411 | 0.9325 | -1.542 | 904.6 | 760.807 | False |
| 13 | 2966 | 92.1224 | 41.6334 | 2.2127 | 0.8921 | 0.895 | -1.4899 | 920.401 | 307.385 | False |
| 14 | 1624 | 72.5441 | 29.7651 | 2.4372 | 0.9119 | 0.8476 | -0.945 | 978.497 | 437.615 | False |
| 15 | | | | | | | | 548.08 | 212.663 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 7 | 192 | 359 | 1753.54 | 96 | 18.2661 | |
| 2 | 14 | 556 | 1343 | 6503.61 | 282 | 23.0624 | 6.9555 |
| 3 | 8 | 363 | 942 | 4290.4 | 197 | 21.7787 | 7.2279 |
| 4 | 16 | 459 | 1235 | 3695.9 | 192 | 19.2495 | 7.4568 |

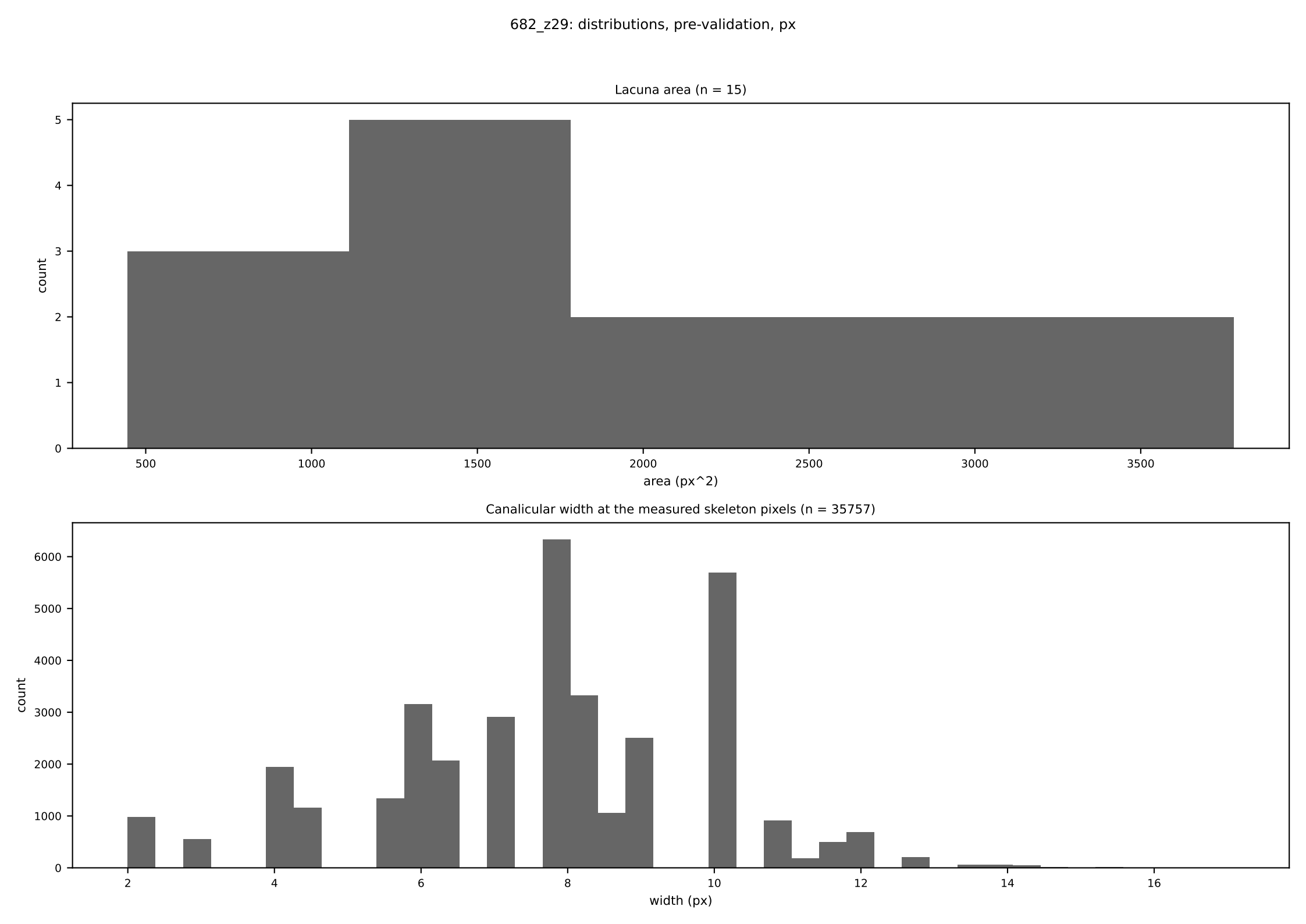
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5 | 20 | 460 | 996 | 3959.72 | 164 | 24.1447 | 6.2493 |
| 6 | 17 | 481 | 950 | 3368.16 | 148 | 22.7579 | 6.9706 |
| 7 | 13 | 392 | 821 | 1236.62 | 60 | 20.6103 | 7.4798 |
| 8 | 16 | 376 | 951 | 1458.23 | 62 | 23.5198 | 6.2347 |
| 9 | 14 | 222 | 673 | 2846.64 | 116 | 24.54 | 6.8502 |
| 10 | 6 | 188 | 481 | 513.162 | 24 | 21.3817 | 6.6465 |
| 11 | 9 | 191 | 480 | 626.831 | 24 | 26.1179 | 8.6113 |
| 12 | 9 | 210 | 648 | 1494.79 | 58 | 25.7723 | 6.5206 |
| 13 | 16 | 445 | 938 | 1579.88 | 62 | 25.482 | 6.7132 |
| 14 | 13 | 272 | 452 | 539.688 | 21 | 25.6994 | 7.0847 |
| 15 | 6 | 245 | 624 | 854.541 | 39 | 21.9113 | 7.0877 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 262.096 | 5607 | 13819 | 0.5809 | 0.5584 | 0.0342429 | 0.0259787 | 2.6708 |
| 2 | 229.522 | 9151 | 23871 | 1 | 0.999 | 0.0607584 | 0.0562607 | 6.0996 |
| 3 | 120.811 | 6134 | 17911 | 1 | 1 | 0.0591783 | 0.0525934 | 6.6219 |
| 4 | 309.279 | 11331 | 27269 | 1 | 1 | 0.0405083 | 0.0452895 | 5.1733 |
| 5 | 235.238 | 9387 | 21888 | 1 | 1 | 0.0490039 | 0.0455044 | 8.502 |
| 6 | 214.167 | 8637 | 19737 | 1 | 1 | 0.0556906 | 0.048133 | 7.9377 |
| 7 | 322.836 | 9940 | 21386 | 0.8402 | 0.7724 | 0.0394366 | 0.0383896 | 4.0268 |
| 8 | 229.48 | 8595 | 22042 | 1 | 1 | 0.0437464 | 0.0431449 | 6.9723 |
| 9 | 156.853 | 6922 | 19429 | 1 | 1 | 0.0320717 | 0.0346389 | 8.9256 |
| 10 | 109.498 | 4245 | 11066 | 1 | 0.9865 | 0.0442874 | 0.0434665 | 5.4796 |
| 11 | 161.817 | 5491 | 13376 | 1 | 1 | 0.0347842 | 0.0358852 | 5.5618 |
| 12 | 147.196 | 6914 | 19480 | 1 | 1 | 0.0303732 | 0.0332649 | 6.1143 |
| 13 | 254.693 | 9728 | 23664 | 1 | 1 | 0.0457442 | 0.0396383 | 6.2821 |
| 14 | 202.693 | 7711 | 16371 | 0.9376 | 0.8055 | 0.0352743 | 0.0276098 | 6.4136 |
| 15 | 47.4558 | 4187 | 11845 | 1 | 1 | 0.0585145 | 0.0526805 | 12.6433 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 85 | 150 | 219.51 | 395.664 | 2 | 6 | 6 |
| 2 | 548 | 1291 | 617.718 | 1522.9 | 12 | 16 | 18 |
| 3 | 319 | 942 | 410.635 | 1048.9 | 8 | 11 | 11 |
| 4 | 428 | 1116 | 519.818 | 1392.83 | 12 | 13 | 16 |
| 5 | 432 | 831 | 505.534 | 1101.01 | 12 | 15 | 14 |
| 6 | 429 | 900 | 539.99 | 1066.98 | 10 | 13 | 15 |
| 7 | 379 | 769 | 440.563 | 924.896 | 10 | 12 | 12 |
| 8 | 358 | 739 | 420.735 | 1071.29 | 10 | 12 | 12 |
| 9 | 199 | 515 | 263.522 | 786.423 | 5 | 7 | 10 |
| 10 | 176 | 392 | 210.338 | 535.576 | 6 | 6 | 6 |
| 11 | 191 | 450 | 221.823 | 534.09 | 5 | 7 | 8 |
| 12 | 157 | 481 | 236.48 | 752.21 | 6 | 7 | 8 |
| 13 | 388 | 744 | 500.718 | 1051.44 | 11 | 17 | 17 |
| 14 | 253 | 366 | 307.522 | 508.86 | 6 | 7 | 7 |
| 15 | 177 | 460 | 273.409 | 706.529 | 4 | 6 | 10 |


682_z29: distributions, pre-validation, px
Lacuna area (n = 15)
count
area (px^2)
0
1
2
3
4
5
500
1000
1500
2000
2500
3000
3500
Canalicular width at the measured skeleton pixels (n = 35757)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16